Computer Networks Study Booklet

Preface

Welcome to the Computer Networks Study Booklet. This document reorganizes scattered classroom notes, textbook chapters, and RFC concepts into core topic families. Instead of memorizing hundreds of disconnected facts or full protocol specifications, this booklet focuses on **pattern recognition, architectural logic, and exam strategy**. It is designed for fast revision, emphasizing how networks are layered, how protocols interact, and how data safely navigates the globe.

Callout Legend:
Exam Lens: How to approach typical exam questions and formulate answers.
Common Trap: Misconceptions and easy mistakes to avoid.
Key Takeaway: The core mental model or summary fact to commit to memory.

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1. Network Foundations and Layered Thinking

When this topic appears: Questions asking to compare fundamental networking paradigms (e.g., how telephones work vs. how the Internet works) or define basic terminology like hosts, end

systems, routers, and links.

Core Ideas

- **Network Edge vs. Core:** The *edge* consists of end systems (hosts like PCs, smartphones, servers). The *core* consists of interconnected routers and switches that forward data.
- **Protocols:** A protocol defines the format, order of messages sent/received, and actions taken on message transmission/receipt.

Feature	Circuit Switching (e.g., Traditional Telecom)	Packet Switching (e.g., The Internet)
Path	Dedicated path reserved for the entire session.	No dedicated path; data routed dynamically.
Resource Allocation	Resources reserved in advance (guaranteed capacity).	Resources shared on demand (statistical multiplexing).
Efficiency	Wasteful if the connection is idle.	Highly efficient use of overall bandwidth.

> **EXAM LENS**
>
> If asked "Why does the Internet use packet switching?", explicitly mention **statistical multiplexing** and **resiliency** (routing around failures).

> **COMMON TRAP**
>
> Don't confuse nodes with links. Nodes are devices (hosts, routers). Links are the physical media connecting them (copper, fiber, wireless).

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2. OSI Model, TCP/IP Model, and Encapsulation

When this topic appears: Almost constantly. Exam questions will ask you to identify which layer a protocol belongs to, or describe the process of a message moving down the stack at the sender and up at the receiver.

Core Ideas

- **Layering:** Breaks down complex communication into manageable, modular pieces. Each layer provides a service to the layer above it.
- **Encapsulation:** As data moves down the stack, each layer adds its own header (and sometimes trailer). Application Data â†’ Transport Segment â†’ Network Datagram (Packet) â†’ Link Frame â†’ Physical Bits.

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Data Unit Name	Key Protocols
7. Application 6. Presentation 5. Session	Application	Message / Data	HTTP, DNS, SMTP
4. Transport	Transport	Segment / Datagram	TCP, UDP
3. Network	Internet	Packet / Datagram	IPv4, IPv6, ICMP
2. Data Link 1. Physical	Network Access / Link	Frame (L2) / Bits (L1)	Ethernet, Wi-Fi (802.11)

> **EXAM LENS**
>
> When describing encapsulation, clearly state the terminology: Layer 4 creates *Segments*, Layer 3 creates *Packets*, Layer 2 creates *Frames*. Using the correct noun proves you know the layer.

> **KEY TAKEAWAY**
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> Routers operate up to Layer 3 (Network). Switches operate up to Layer 2 (Data Link). Hubs are Layer 1 (Physical) devices.

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3. Physical Layer and Data Link Layer Essentials

When this topic appears: Discussions on MAC addresses, local area networks (LANs), Ethernet, collision detection, and basic switching logic.

Core Ideas

- **MAC Address:** 48-bit physical address, usually burned into the NIC. Represented in hex (e.g., 00:1A:2B:3C:4D:5E). Used for local delivery on the same link/subnet.
- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):** "Listen before you talk. If someone interrupts, stop talking, wait a random time, and try again." Primarily historical for wired Ethernet (switches made it largely obsolete), but the concept is highly testable.
- **ARP (Address Resolution Protocol):** Translates a known IP address to an unknown MAC address. "Who has IP X.X.X.X? Tell Y.Y.Y.Y."

Device	Layer	Behavior
Hub	1 (Physical)	Broadcasts incoming bits to all other ports. Single collision domain.

Switch	2 (Data Link)	Learns MAC addresses (MAC Table). Forwards frames only to the destination port.
Router	3 (Network)	Routes packets between different networks based on IP addresses.

> **COMMON TRAP**
>
> A switch does not use IP addresses to forward traffic. It looks exclusively at the destination MAC address in the Ethernet frame.

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4. IP Addressing, Subnetting, and Routing Logic

When this topic appears: Calculating subnets, distinguishing between routing and forwarding, and comparing IPv4 vs IPv6.

Core Ideas

- **IPv4 vs IPv6:** IPv4 is 32-bit (approx 4.3 billion addresses). IPv6 is 128-bit (designed to fix IP exhaustion).
- **Routing vs. Forwarding:**
  - *Routing (Control Plane):* The brain. Algorithms (OSPF, BGP) computing the best path to a destination and building the routing table.
  - *Forwarding (Data Plane):* The muscle. Moving a packet from an input port to the correct output port based on the forwarding table.

Quick Subnetting Method

Subnetting borrows bits from the host portion to create network subnets. Given a CIDR notation (e.g., /26):

1. **Identify Network Bits:** /26 means 26 bits for the network, 6 bits left for hosts (32 - 26 = 6).
2. **Calculate Hosts per Subnet:** Formula is `2^h - 2`. Here, 2^6 - 2 = 64 - 2 = 62 usable hosts. (Subtract 2 for Network ID and Broadcast ID).
3. **Calculate Subnet Mask:** 26 bits = 255.255.255.192.

> **EXAM LENS**
>
> For subnetting questions, immediately write down powers of 2 (128, 64, 32, 16, 8, 4, 2, 1) in your scratch paper. Remember that the "magic number" (block size) is 256 - the interesting octet of the subnet mask.

> **COMMON TRAP**

> Never assign the Network Address (all host bits are 0) or the Broadcast Address (all host bits are 1) to a PC or router interface.

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5. Transport Layer: UDP, TCP, Reliability, Flow & Congestion Control

When this topic appears: The most heavily tested area. Expect detailed questions on TCP's 3-way handshake, state machines, and the mechanics of congestion control.

Core Ideas

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (3-way handshake).	Connectionless (fire and forget).
Reliability	Guarantees delivery, ordering, and data integrity.	Best effort. Packets can be lost or out of order.
Overhead	High (20-byte header, state tracking).	Low (8-byte header).
Use Case	Web (HTTP), Email, File Transfer.	Streaming, DNS, VoIP, Gaming.

TCP Deep Dive in Plain English

- **3-Way Handshake:** SYN â†’ SYN-ACK â†’ ACK. Establishes sequence numbers to track data.
- **Reliable Delivery & Sliding Window:** TCP sends a block (window) of data. The receiver sends ACKs. If an ACK isn't received before a timer expires, TCP retransmits. The window "slides" forward as data is acknowledged.
- **Flow Control vs. Congestion Control:**
  - *Flow Control:* Protecting the **receiver**. The receiver advertises its buffer size (Receive Window) so the sender doesn't overwhelm it.
  - *Congestion Control:* Protecting the **network**. The sender uses Slow Start (exponential growth) and Congestion Avoidance (linear growth) to avoid causing traffic jams in core routers.

> **EXAM LENS**
>
> If asked to explain TCP Congestion Control, mention the phases explicitly: **Slow Start** (cwnd doubles every RTT), **Congestion Avoidance** (cwnd increases by 1 MSS every RTT), and **Fast Retransmit/Fast Recovery** (triggered by 3 duplicate ACKs).

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6. Application Layer: DNS, HTTP/HTTPS, DHCP, Email Basics

When this topic appears: Questions tracing the life of a web request ("What happens when you type www.google.com?").

Core Ideas

- **DNS (Domain Name System):** The internet's phonebook. Converts human-readable names (example.com) to IP addresses. Architecture is hierarchical: Root servers â† ' Top-Level Domain (TLD) servers â†' Authoritative servers.
- **HTTP/HTTPS:** Request-Response protocol. Stateless by default (uses cookies to remember state). HTTPS adds TLS/SSL for encryption, authentication, and integrity.
- **DHCP (Dynamic Host Configuration Protocol):** Assigns IPs dynamically. Flow is DORA: **D**iscover (broadcast), **O**ffer (unicast/broadcast), **R**equest (broadcast), **A**cknowledge (unicast/broadcast).
- **Email:** SMTP pushes email from client to server and server to server. IMAP/POP3 pulls email from the server to the client.

> **KEY TAKEAWAY**
>
> Before an HTTP GET request can even be sent, the client must resolve the IP via DNS, then establish a TCP connection (3-way handshake). If HTTPS, a TLS handshake follows TCP. Only then does the HTTP request travel.

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7. Performance, Delay, Throughput, Errors, and Security Basics

When this topic appears: Calculating total delay across a network link or identifying the bottleneck in a connection.

Understanding Network Delay (The 4 Types)

1. **Processing Delay:** Time a router takes to examine the header and determine where to direct the packet. (Usually negligible).
2. **Queueing Delay:** Time the packet sits in the router's buffer waiting for the link to become available. Highly variable based on network congestion.
3. **Transmission Delay:** Time to push all the packet's bits onto the wire. *Formula: L/R* (Length of packet in bits / Rate of link in bps).
4. **Propagation Delay:** Time for a single bit to travel from point A to point B across the medium. *Formula: d/s* (Distance / Speed of light in the medium).

Total Delay = Processing + Queueing + Transmission + Propagation.

Throughput

Throughput is the actual rate at which data is transferred. It is constrained by the **bottleneck link**â€”the slowest link in the end-to-end path.

Security Basics

- **Confidentiality:** Only sender/receiver can understand message (Encryption).
- **Integrity:** Message hasn't been tampered with (Hashes, Checksums).
- **Authentication:** Confirming sender identity (Digital Certificates, RSA).

> **EXAM LENS**
>
> Do not confuse Transmission Delay with Propagation Delay. Transmission depends on packet size and bandwidth. Propagation depends strictly on physical distance and medium speed.

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8. Final Revision Toolkit

Essential Formulas

- **Usable Hosts per Subnet:** `2^h - 2` (where h is the number of host bits).
- **Transmission Delay:** `L / R` (Packet length L in bits divided by Link bandwidth R in bps).
- **Propagation Delay:** `d / s` (Distance d divided by propagation speed s).
- **Utilization (Stop-and-Wait):** `(L/R) / (L/R + 2*Propagation)`.

Protocol Port Cheat Sheet

- **FTP:** 20, 21
- **SSH:** 22
- **Telnet:** 23
- **SMTP:** 25
- **DNS:** 53 (UDP mostly, TCP for zone transfers)
- **DHCP:** 67, 68
- **HTTP:** 80
- **HTTPS:** 443

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9. Practice Problems

Problem 1: Protocol Stack Recognition

A user requests a web page over a Wi-Fi network. List the protocol stack (from Application down to Physical) involved in the very first data frame leaving the user's laptop containing the HTTP request. Assume DNS resolution has already occurred.

Problem 2: Subnetting

You are given the IP address 192.168.1.0/24 and need to create at least 5 subnets, each supporting at least 25 hosts. What is the new subnet mask, and what is the range of the first usable subnet?

Problem 3: Delay Calculation

Host A sends a 10,000-bit packet to Host B over a single link. The link distance is 2,000 km, propagation speed is 2 Ã— 10^8 m/s, and link bandwidth is 1 Mbps. Calculate the Transmission Delay and Propagation Delay.

Problem 4: TCP Congestion Control

A TCP connection's congestion window (cwnd) is currently 8 MSS. A timeout occurs. What happens to the ssthresh (slow start threshold) and cwnd values immediately after the timeout?

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10. Model Solution Sketches

Solution 1 Sketch:

- Application: HTTP
- Transport: TCP (Segment)
- Network: IPv4/IPv6 (Packet)
- Data Link: 802.11 / Wi-Fi MAC (Frame)
- Physical: 802.11 RF Signals (Bits)

Solution 2 Sketch:

- Need 5 subnets â† Requires 3 bits (2^3 = 8 subnets).
- New CIDR: /24 + 3 = /27. Mask: 255.255.255.224.
- Hosts remaining: 32 - 27 = 5 bits. 2^5 - 2 = 30 usable hosts (meets the â‰¥25 requirement).
- First subnet block: 0 to 31.
- First usable range: 192.168.1.1 to 192.168.1.30.

Solution 3 Sketch:

- Transmission Delay = L / R = 10,000 bits / 1,000,000 bps = 0.01 seconds (10 ms).
- Propagation Delay = d / s = 2,000,000 meters / 200,000,000 m/s = 0.01 seconds (10 ms).

Solution 4 Sketch:

- Upon a timeout (severe congestion):
  - `ssthresh` is set to half the current cwnd: 8 / 2 = 4 MSS.
  - `cwnd` drops aggressively back to 1 MSS.

- The sender enters the Slow Start phase.

Disclaimer: This booklet is designed as a study and revision aid for Computer Networks examinations. It simplifies complex technical standards for pedagogical purposes and is not a substitute for complete RFC specifications or vendor documentation.